

- Q.26 Describe working of contact condenser.
- Q.27 Differentiate between fire tube and water tube boilers.
- Q.28 Write construction of Babcock and Wilcox boiler.
- Q.29 Explain evaporation capacity and steam economy.
- Q.30 Draw diagram of falling film evaporator.
- Q.31 Explain forced circulation evaporator.
- Q.32 Classify various types of evaporators.
- Q.33 Explain working of cupola furnace.
- Q.34 Discuss various types of evaporators.
- Q.35 Write applications of furnaces in industry.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe construction and working of shell and tube condenser with diagram.
- Q.37 Write a short note on boilers used in industries. Also give construction and working of Lankashire boiler.
- Q.38 What is LMTD. Derive an expression for LMTD for counter current flow.

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Roll No.

4th Sem /
Chemical Engg., Chem (Pulp & Paper Tech.)
Sub : Heat Transfer – II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Condensation of steam on a cold surface is an example of:
- Radiation
 - Natural convection
 - Phase change heat transfer
 - Mass transfer
- Q.2 Film wise condensation occurs when surface is:
- Oily
 - Clean
 - Rough
 - Insulated
- Q.3 Unit of overall heat transfer coefficient is:
- W
 - W/m²K
 - J/kg
 - m²K/W
- Q.4 Cross flow heat exchanger is mainly used in:
- Condensers
 - Boilers
 - Radiators
 - furnaces

- Q.5 Fouling reduces:
- a) Heat transfer rate b) Pressure
 - c) Temperature d) Area
- Q.6 Condenser used in power plants is generally:
- a) Contact condenser b) Surface condenser
 - c) Jet condenser d) Spray condenser
- Q.7 Babcock and Wilcox boiler is a:
- a) Fire tube boiler b) Package boiler
 - c) Electric boiler d) Water tube boiler
- Q.8 Falling film evaporator works at:
- a) High pressure b) Vacuum only
 - c) Atmospheric pressure d) Low pressure
- Q.9 Steam economy is defined as:
- a) Steam used per kg evaporation
 - b) Kg evaporation per kg steam
 - c) Heat supplied
 - d) Heat rejected
- Q.10 Reverberatory furnace uses:
- a) Direct fuel contact
 - c) Electric heating
 - c) Reflected heat
 - d) Induction heating

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is heat exchanger?
- Q.12 Define LMTD.
- Q.13 What is condenser efficiency?
- Q.14 Name one water tube boiler?
- Q.15 What is evaporation rate?
- Q.16 Define boiling.
- Q.17 What is furnace?
- Q.18 Name one industrial evaporator.
- Q.19 What is heat flux?
- Q.20 Define overall heat transfer coefficient.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain different types of condensation.
- Q.22 Discuss about boiling heat transfer.
- Q.23 What is individual and overall heat transfer coefficient.
- Q.24 Explain construction of double pipe heat exchanger.
- Q.25 What is fouling factor?